

May 27th 2024

Graphene synthesis project update

1. Biochar synthesis

- Overall seven syntheses finished with hemp fibers from BAFA neu GmbH
 - three syntheses done with 0.02M sulfuric acid
 - first product not representative, not used further
 - second and third batch converted (see below)
 - four syntheses done with 0.10M sulfuric acid
 - very similar observations and outputs
 - products of these have been combined/homogenized
 - part of the combined lots has been converted
- Canadian Rockies Hemp arrived and cleared customs
 - seven synthesis batches finished (11.2 g product in total)
- (Re)-Activation of larger vessel in progress

2. Biochar analysis

- No samples submitted yet

3. Graphene synthesis

- two experiments finished with steep heating ramp (20-30 °C/min)
 - biochar ground manually for first experiment
 - laboratory mill used for second experiment
 - products not yet characterized
 - no serious handling issues
- two experiments performed with much gentler heating ramp (3 °C/min)
 - better handling but no other apparent difference

4. Graphene analysis

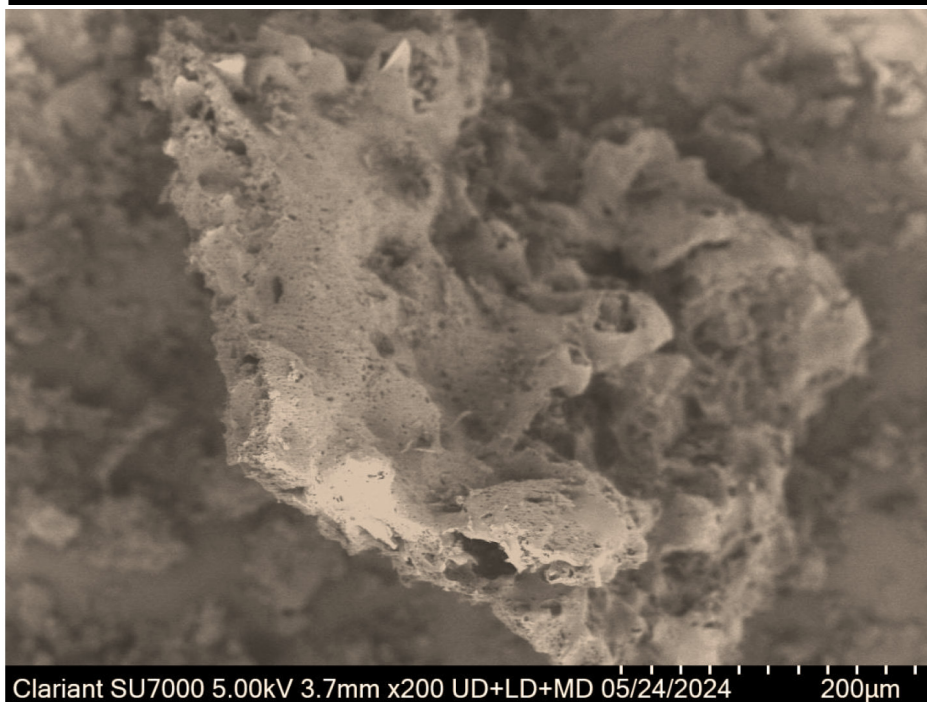
- First samples (TB1133, Sigma-Aldrich Reference) sent out for Raman / SEM
 - SEM Results for TB1133 below
- Very promising discussion with one potential service provider (TEM, BET), expecting offer and NDA draft

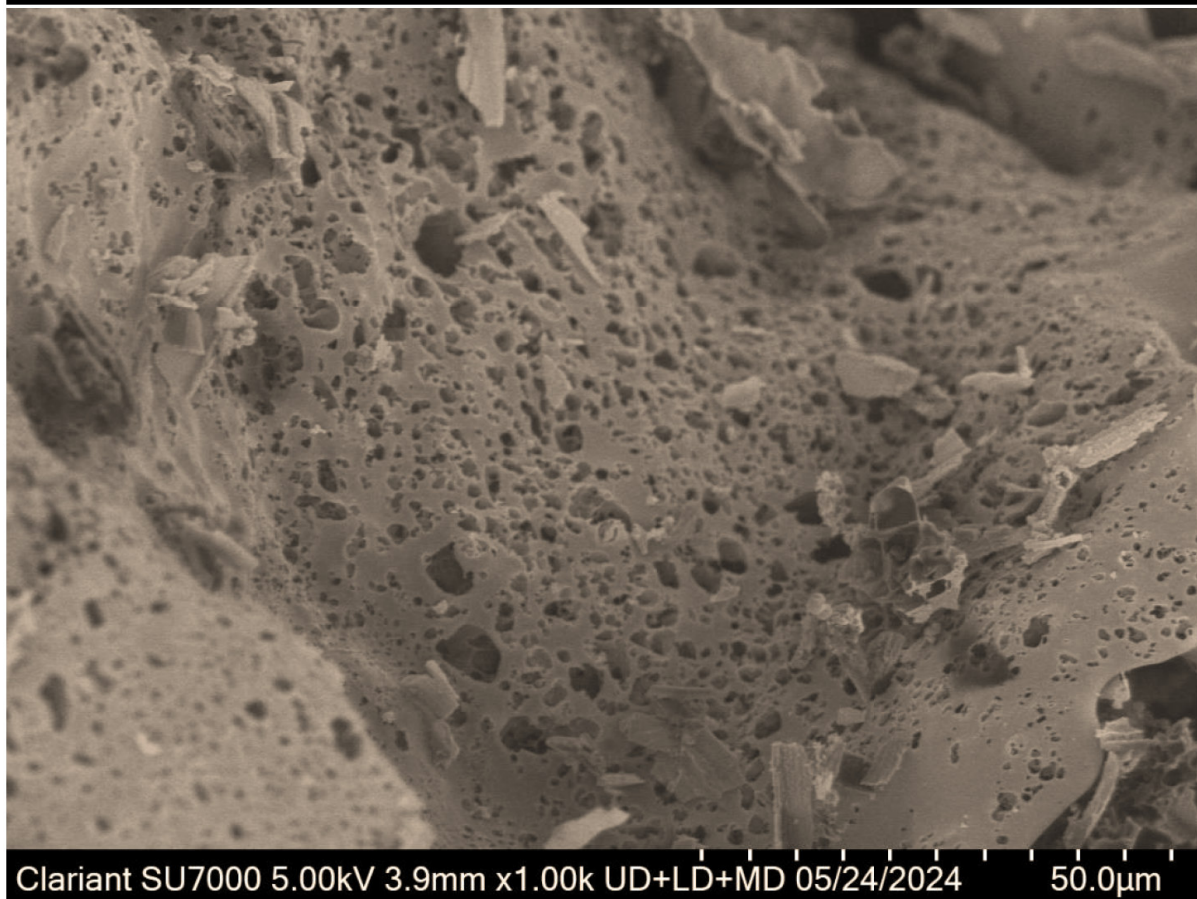
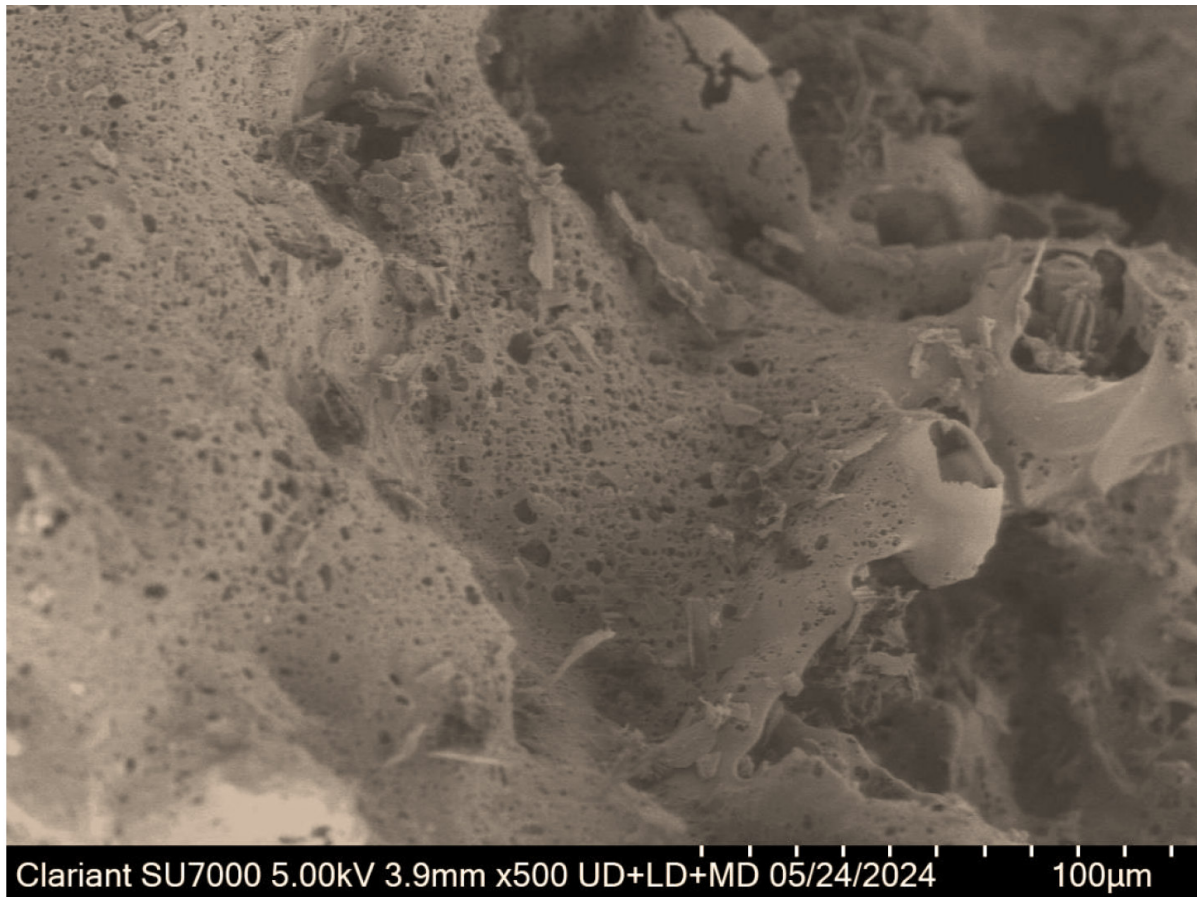
5. Currently open questions

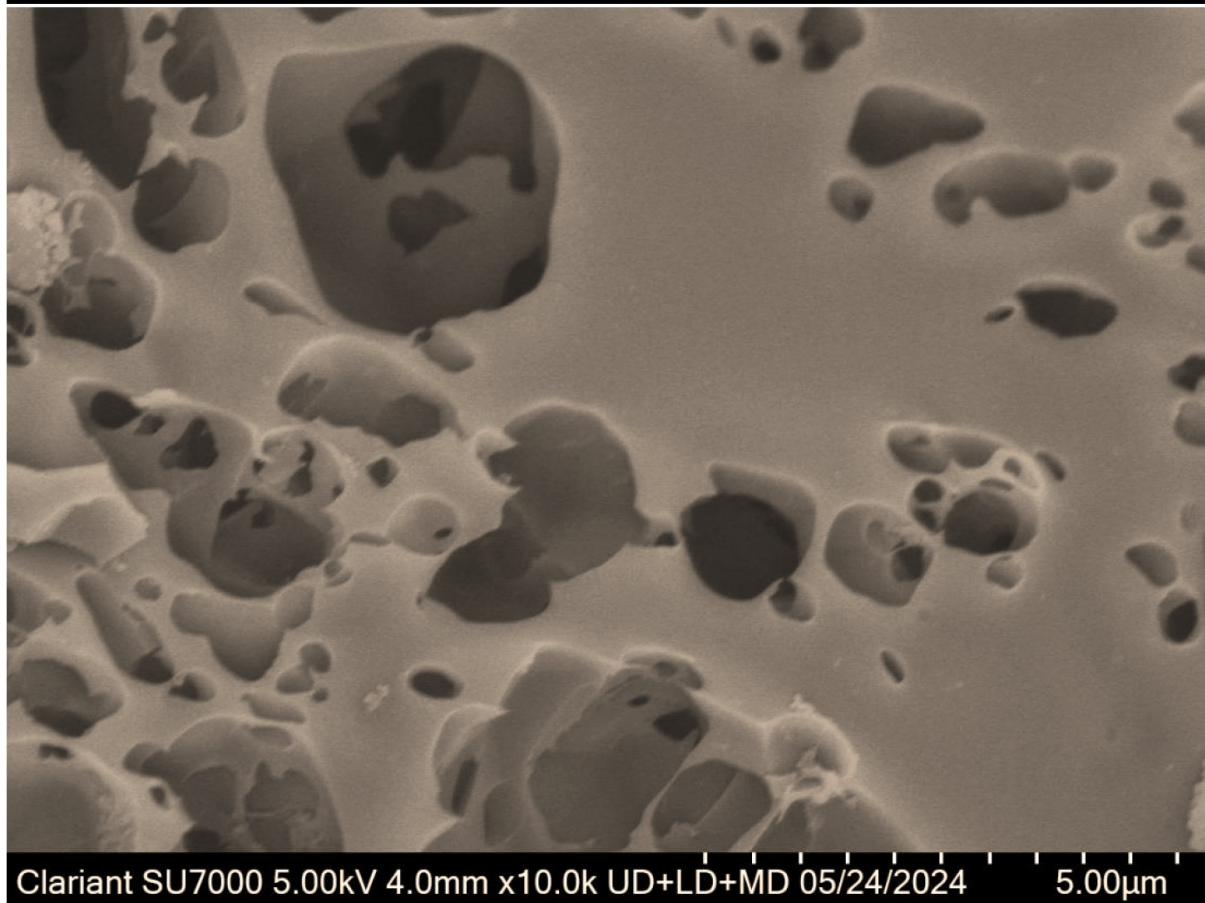
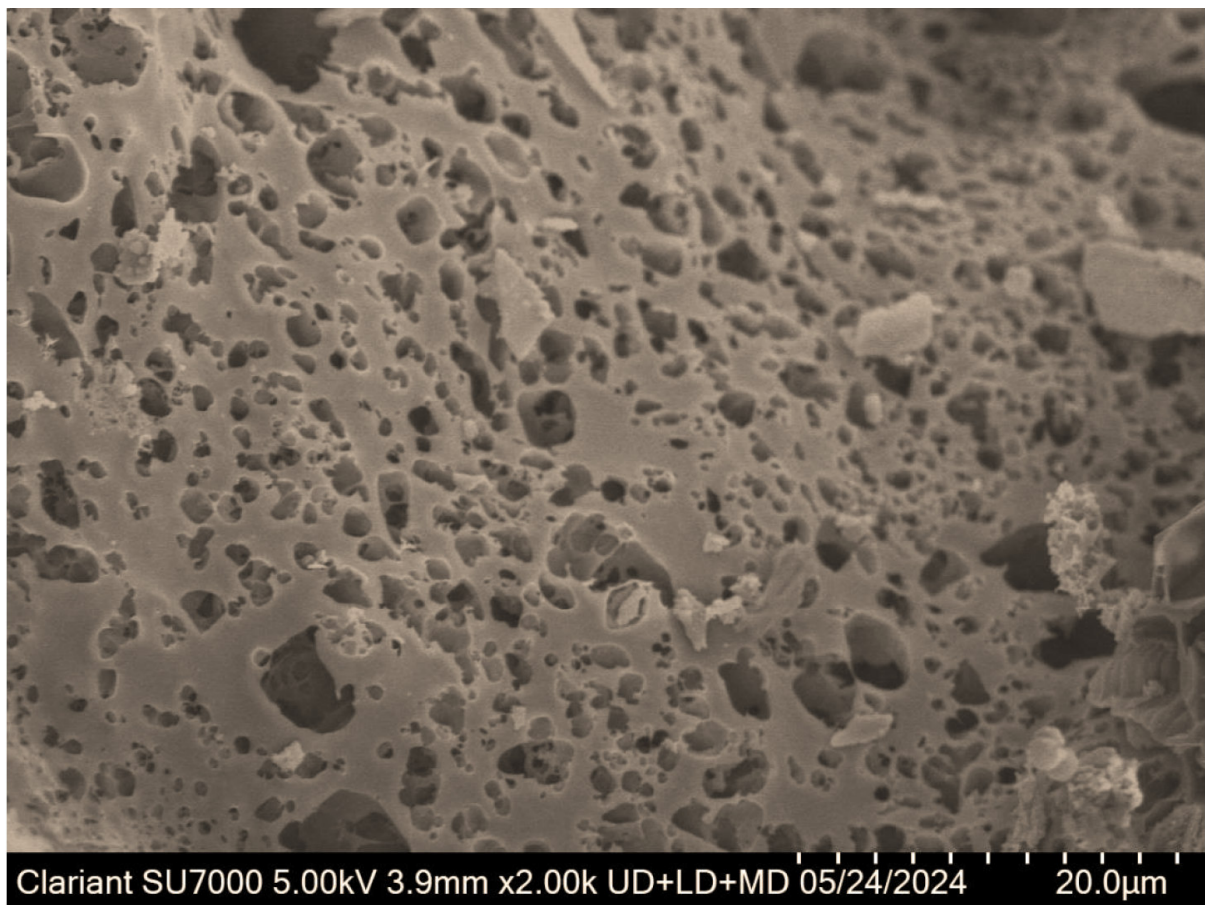
- What type of reaction vessel has the hydrothermal synthesis been done in? Glas? Glas-lined/Enamel? Steel? PTFE?
 - Is it known whether noteworthy corrosion should be expected in steel?
 - Is it known whether iron ions from corrosion have an impact on product quality?

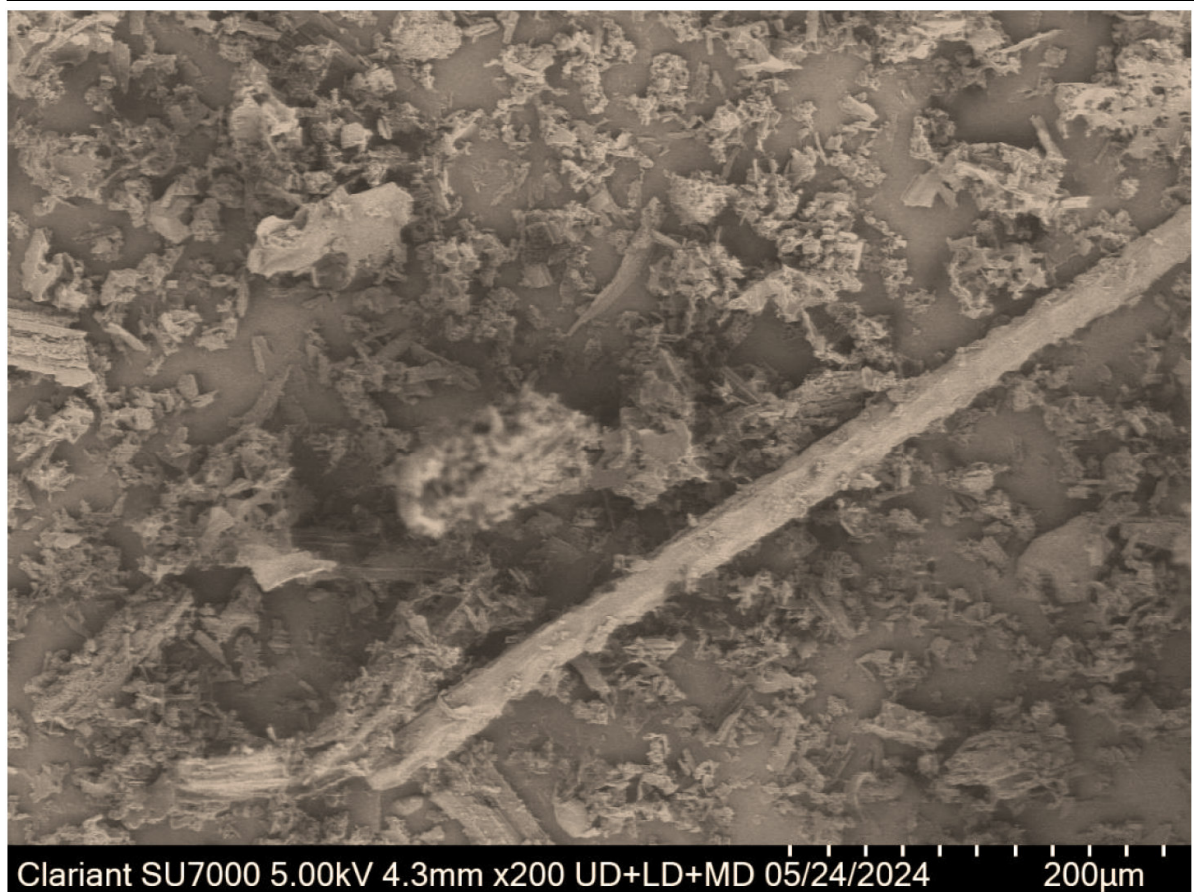
6. Analytical data

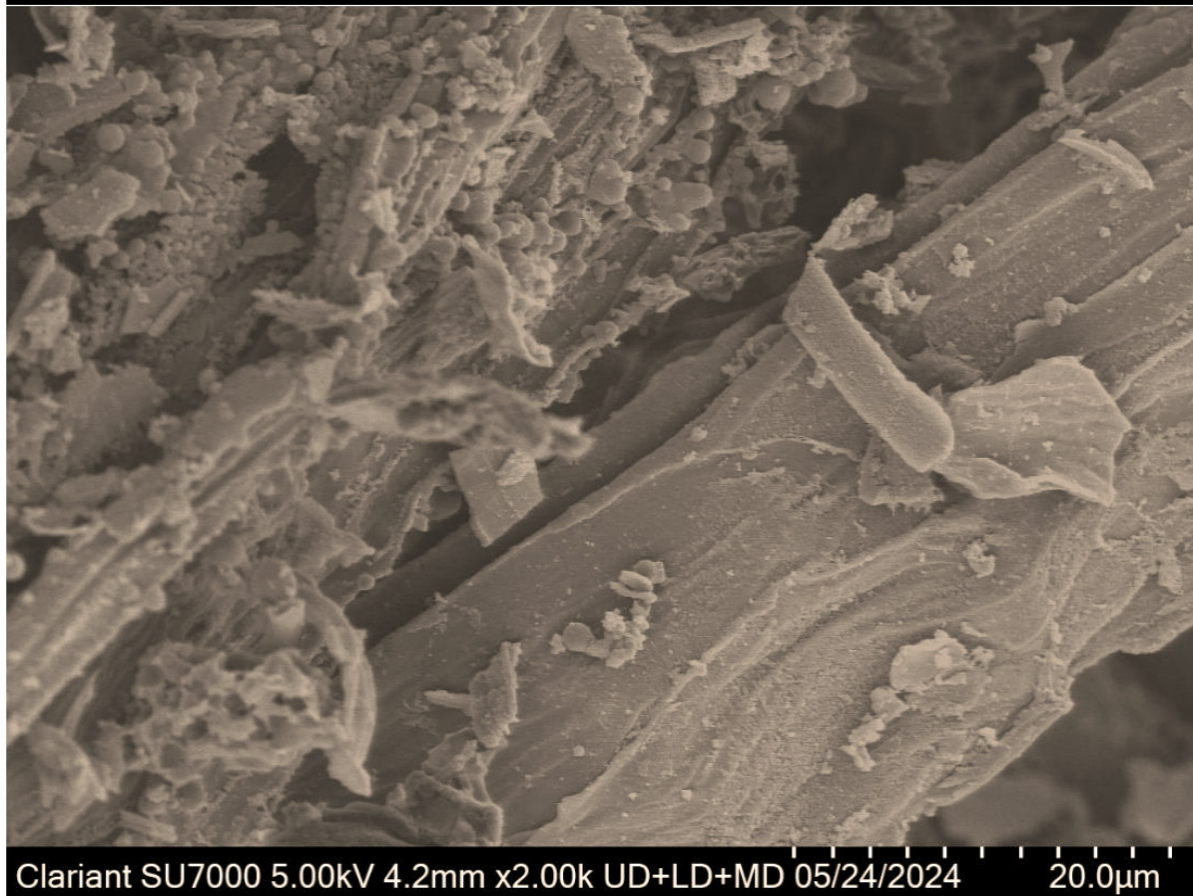
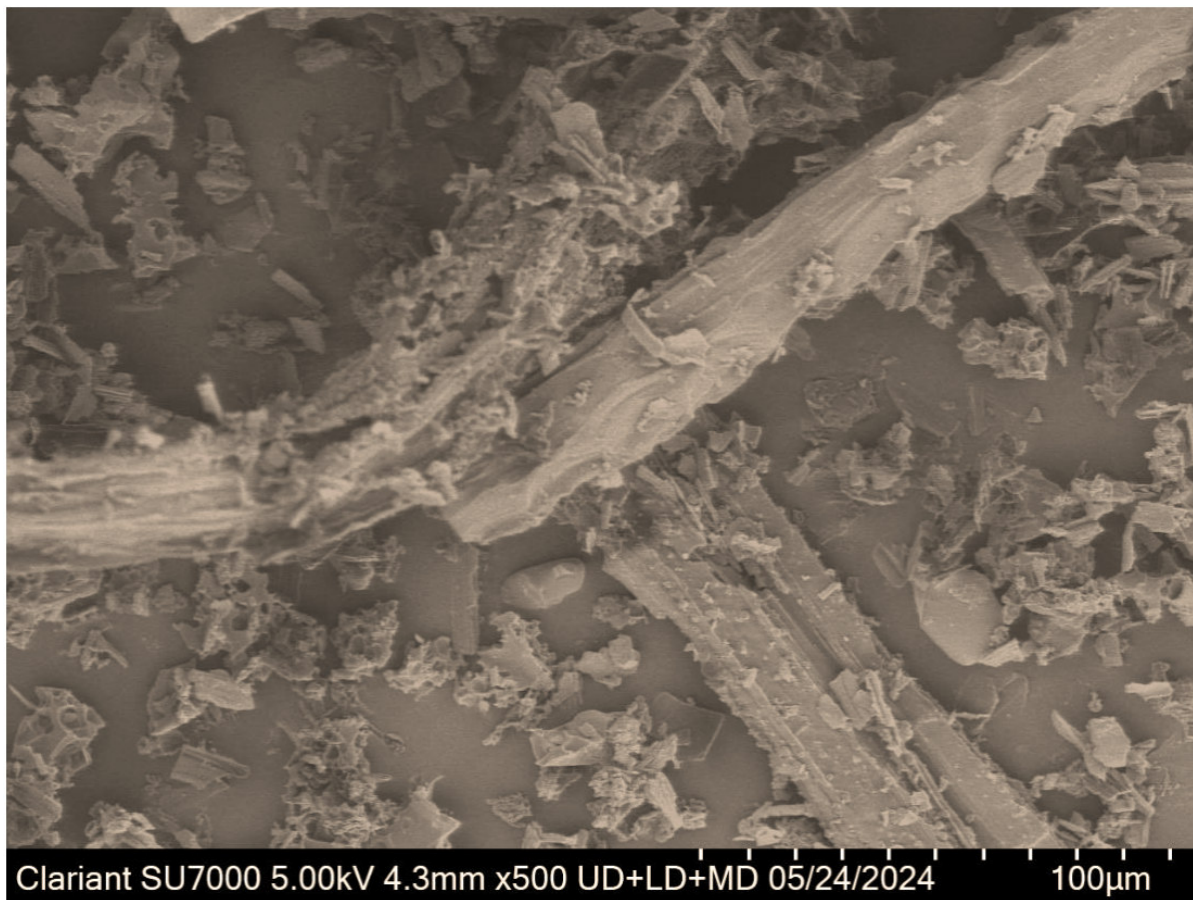
TB1133 - SEM

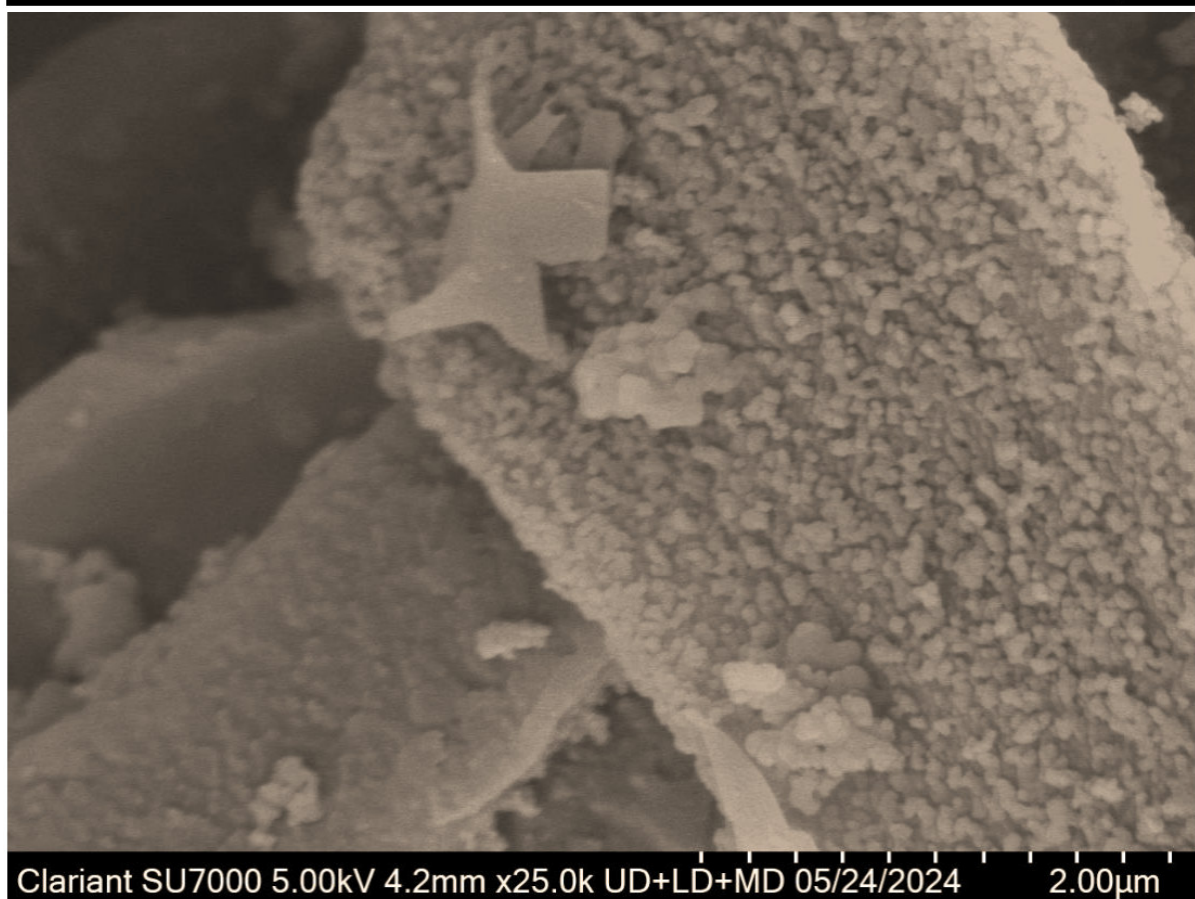
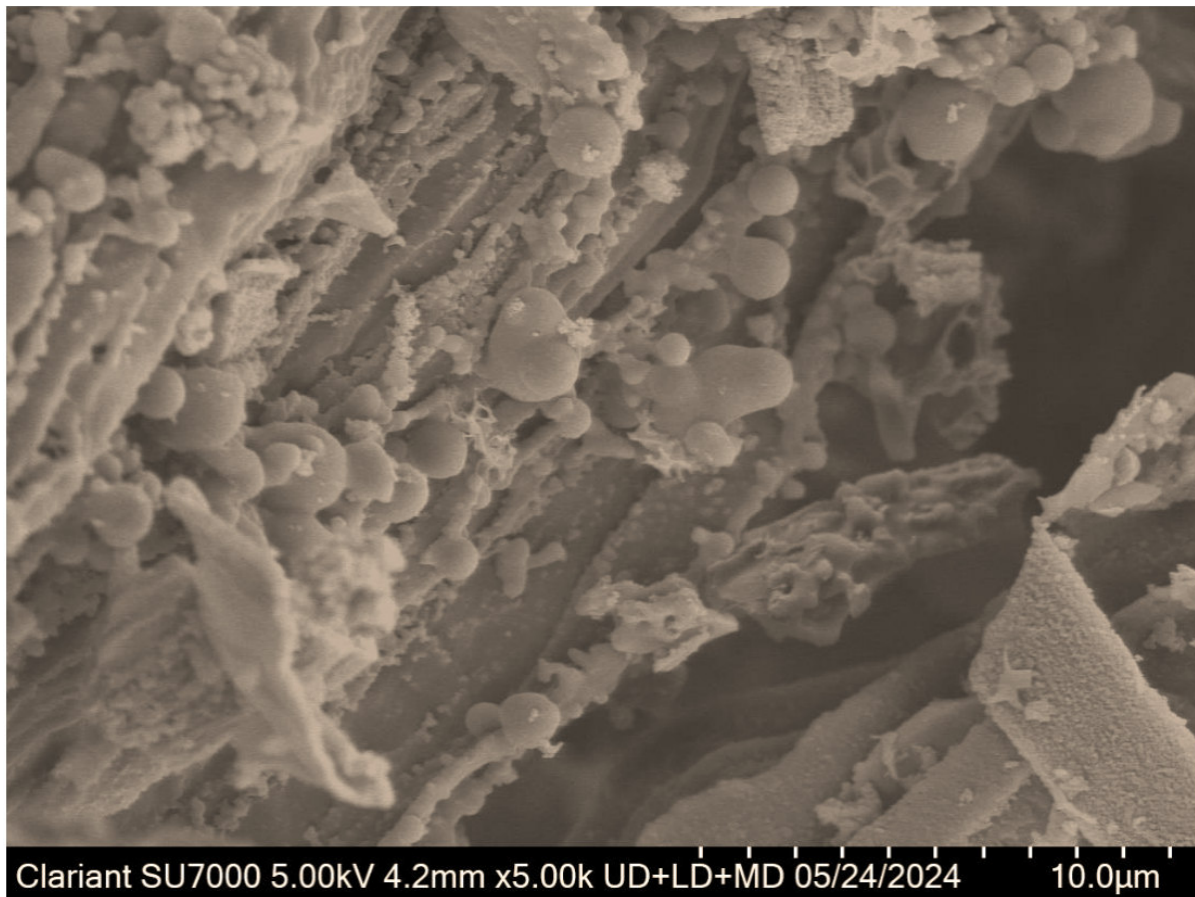


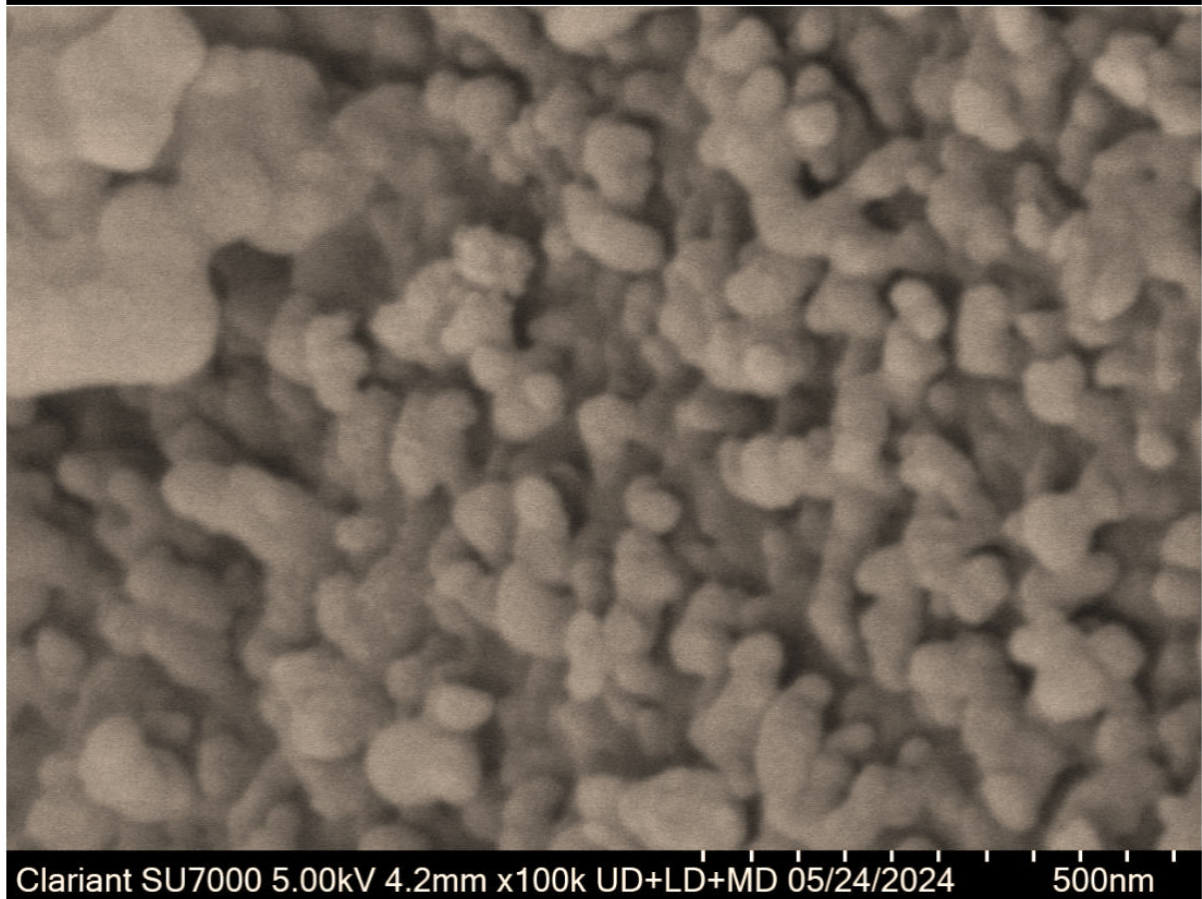
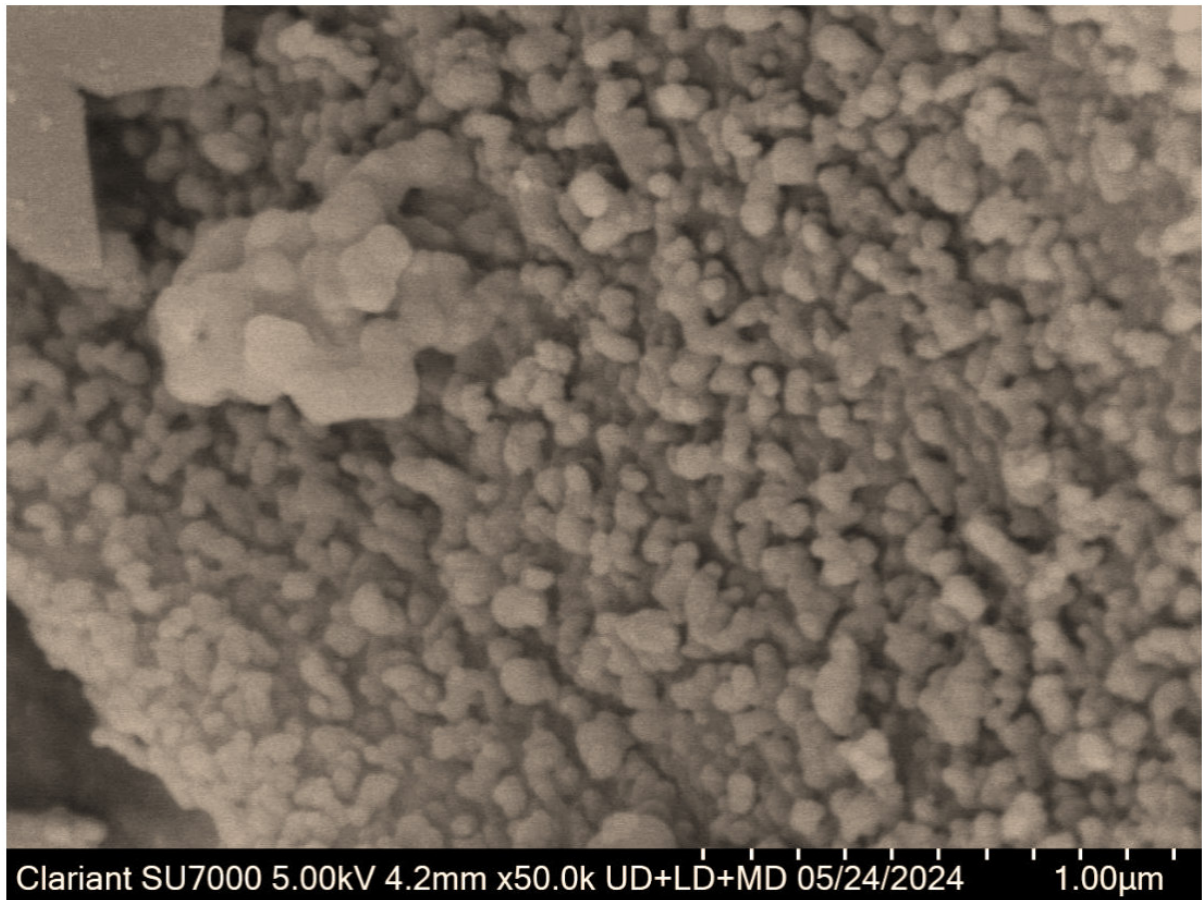




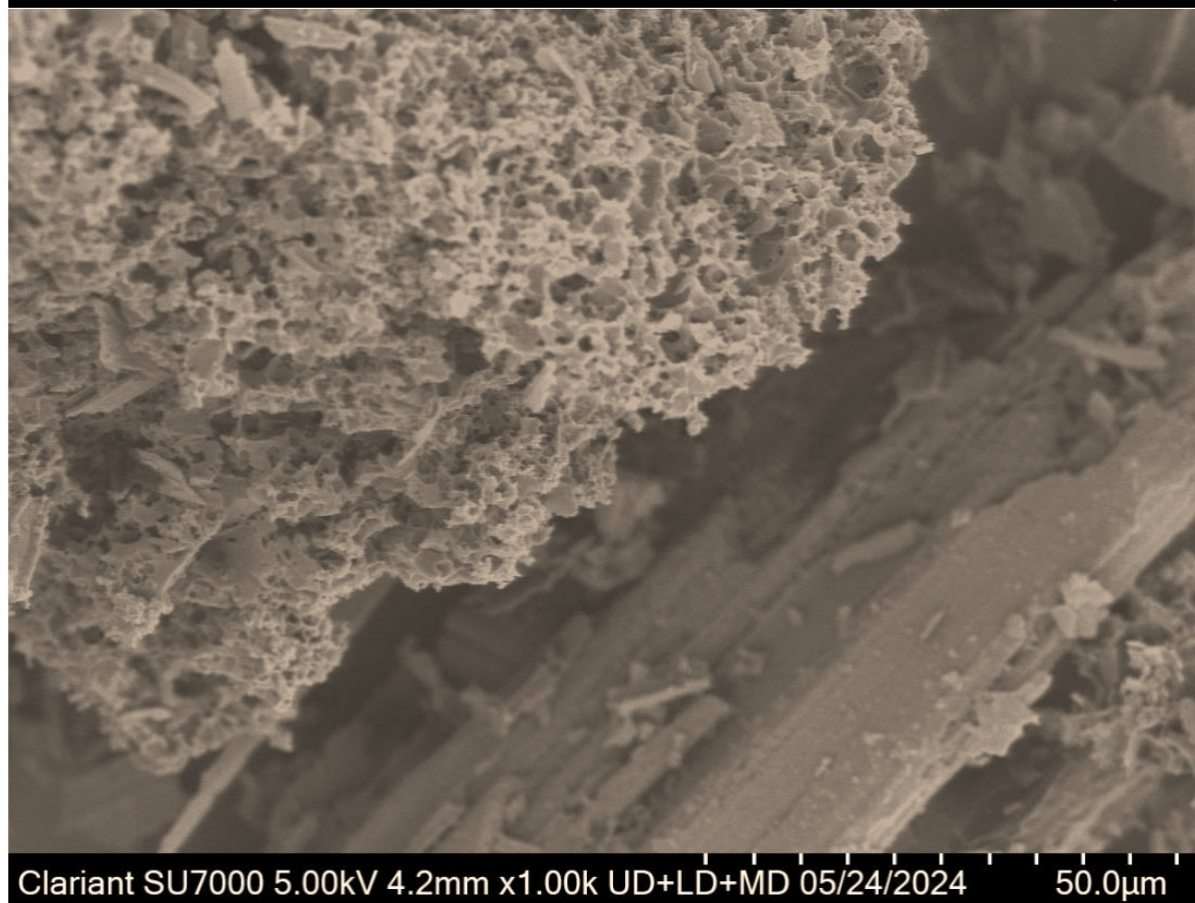
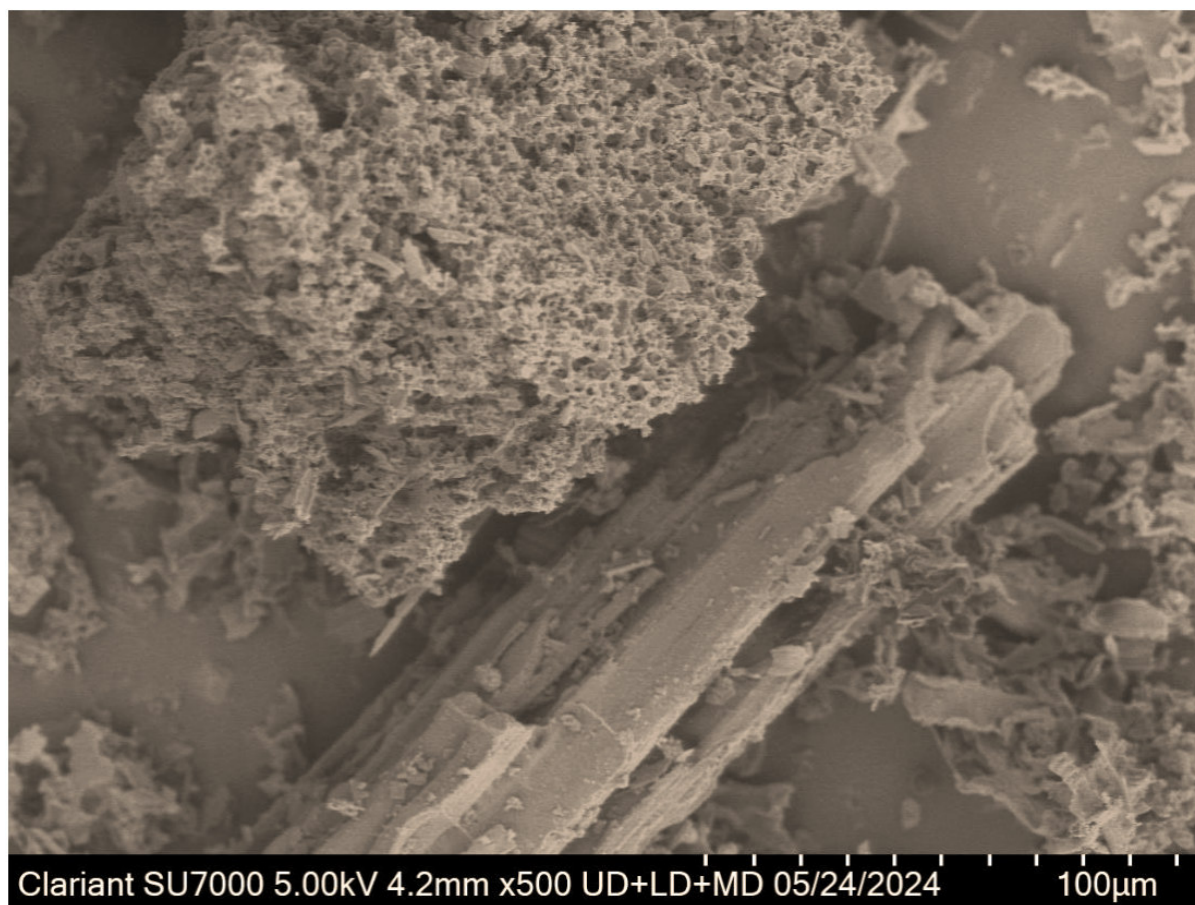


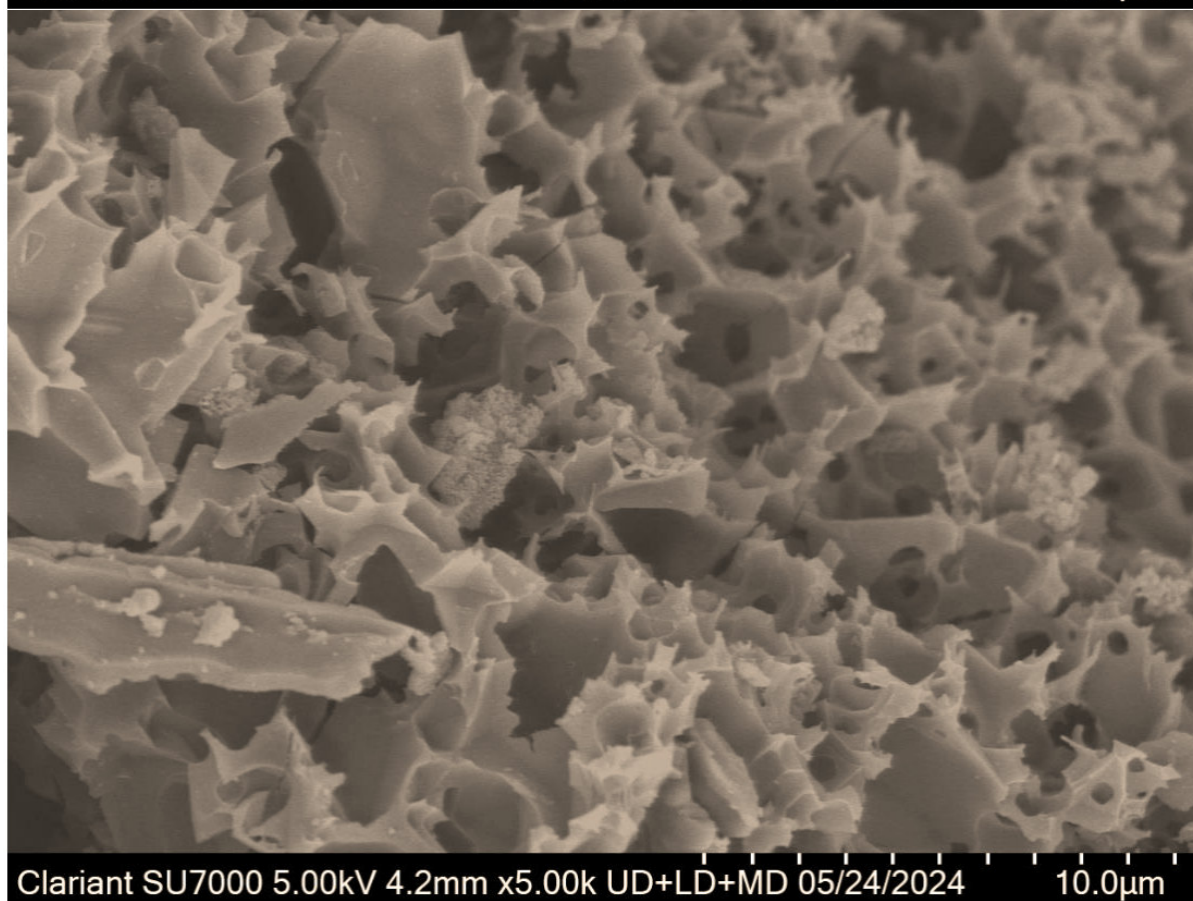
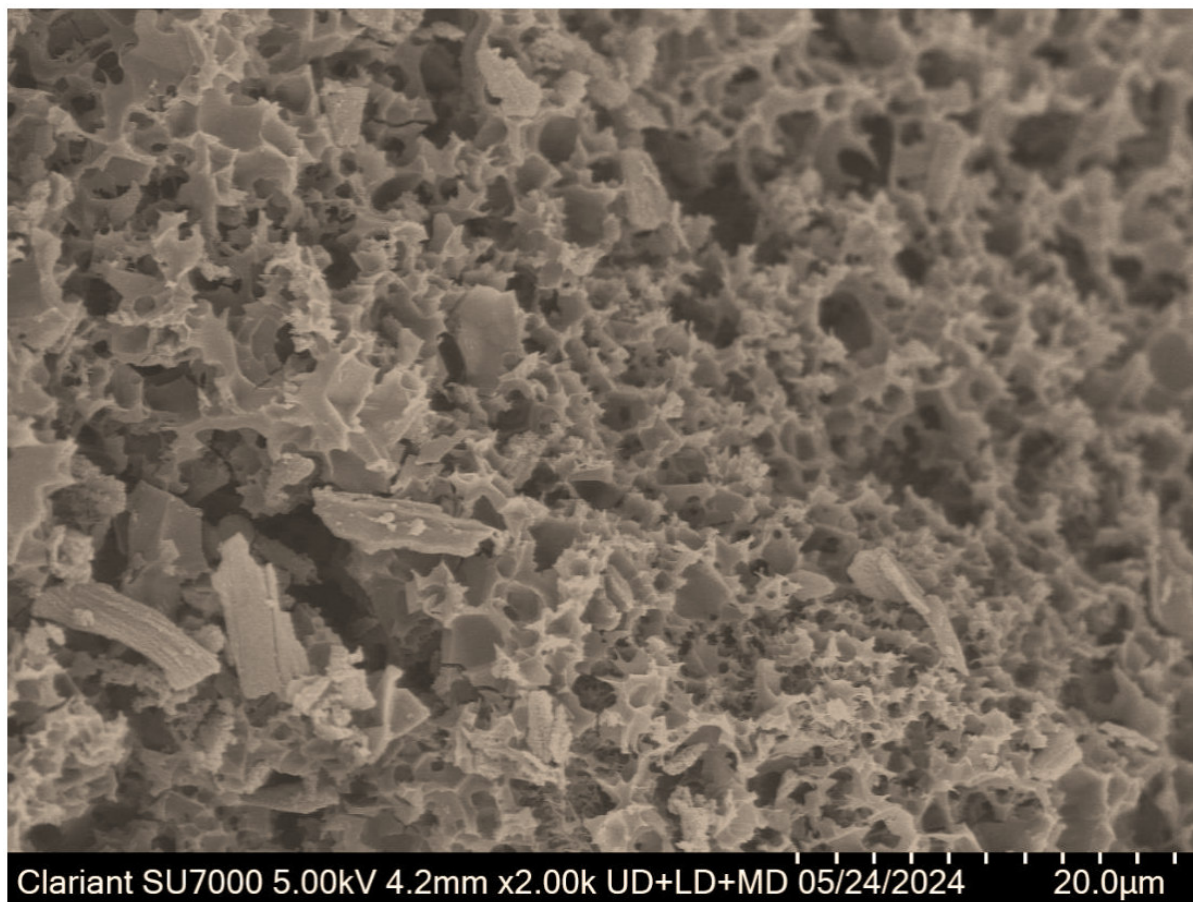


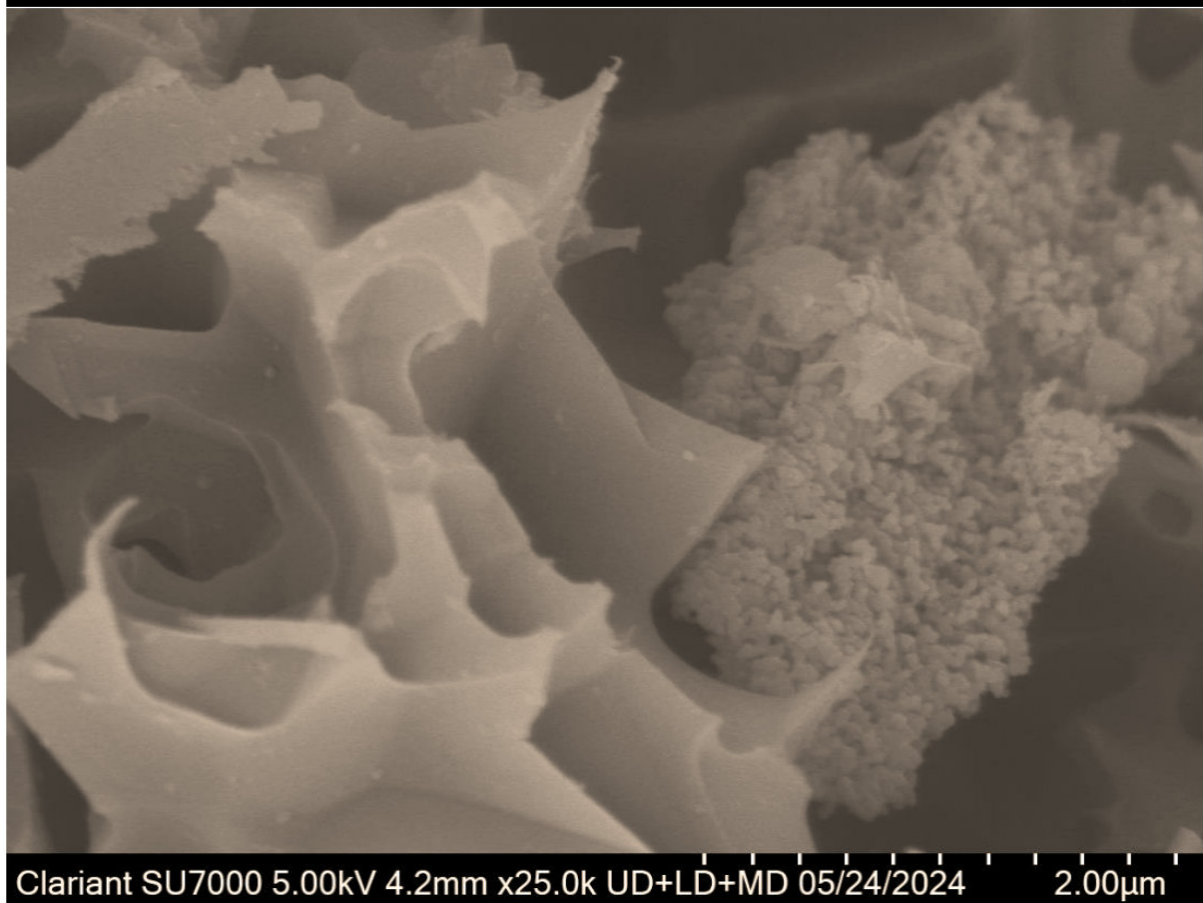
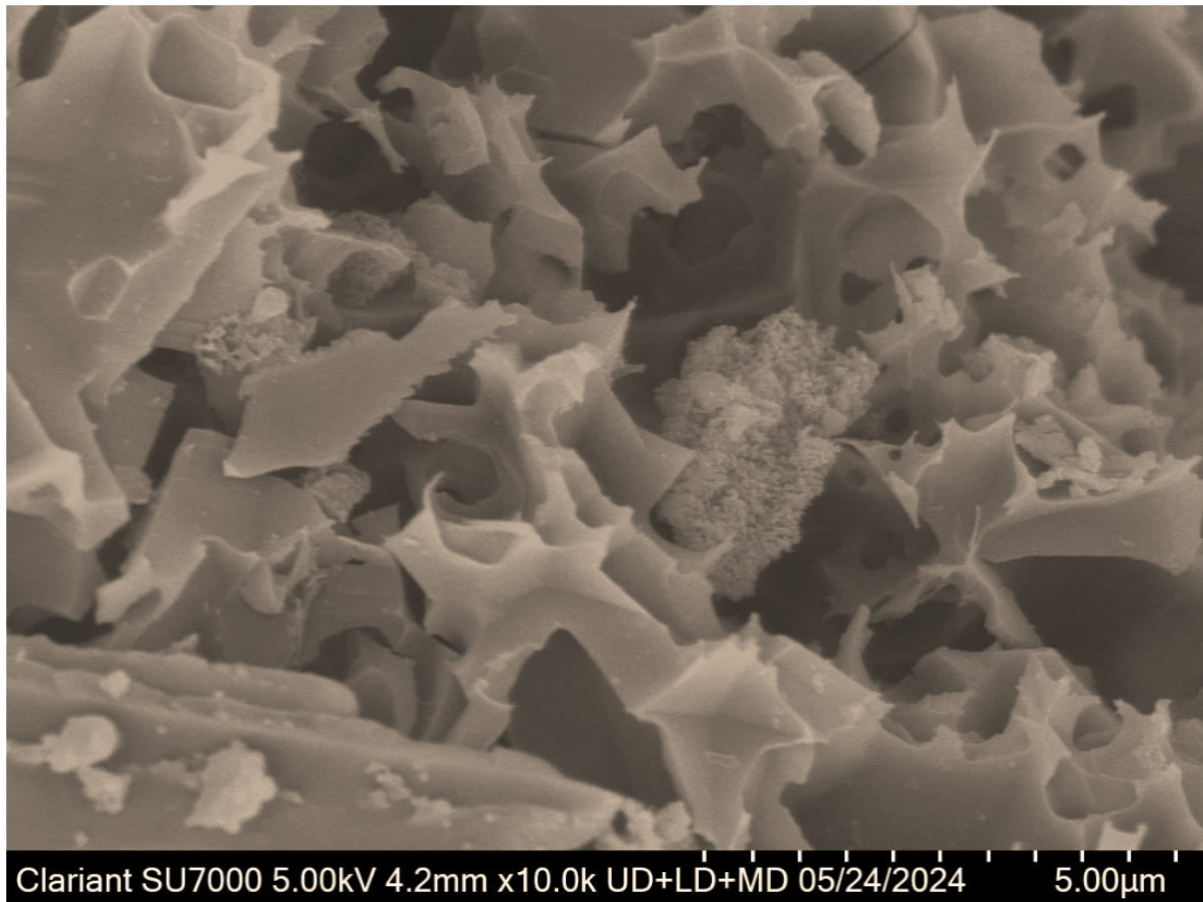


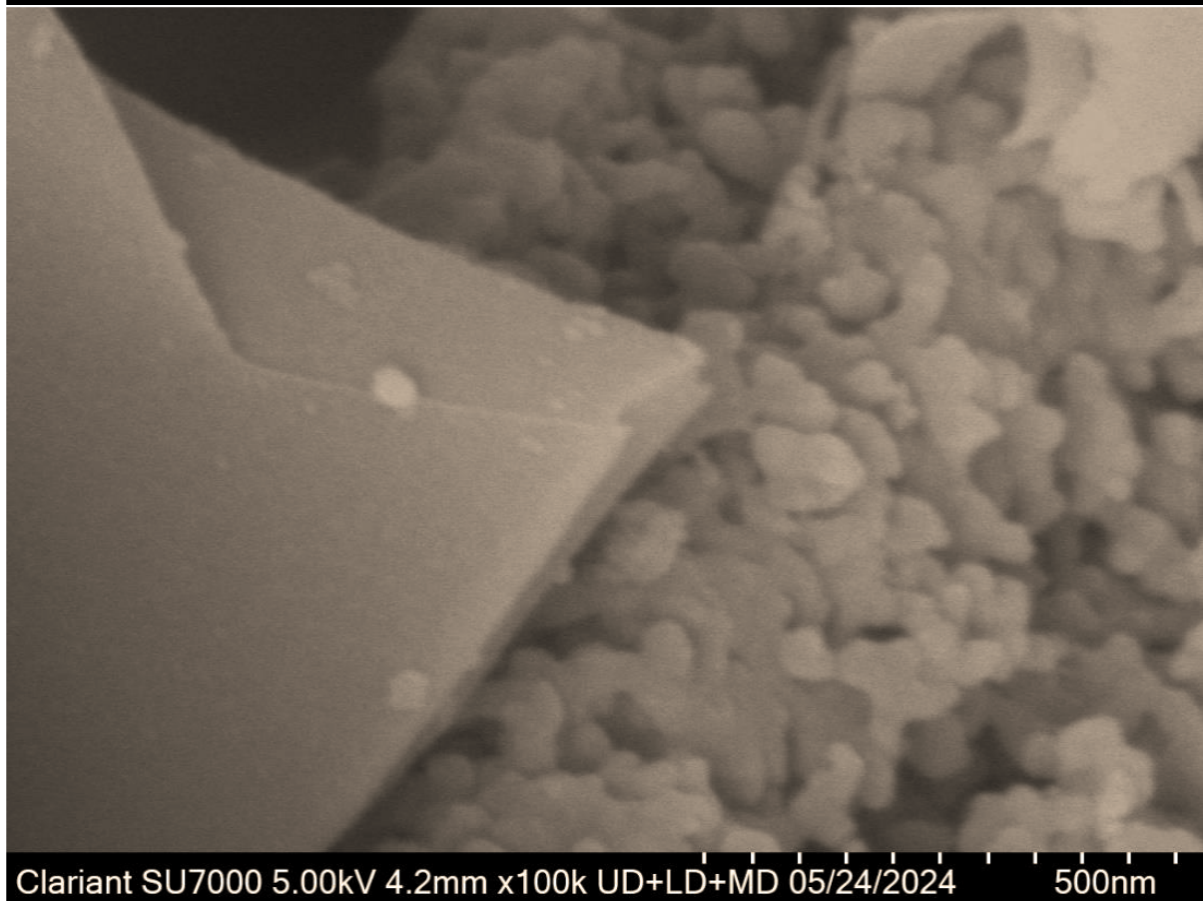
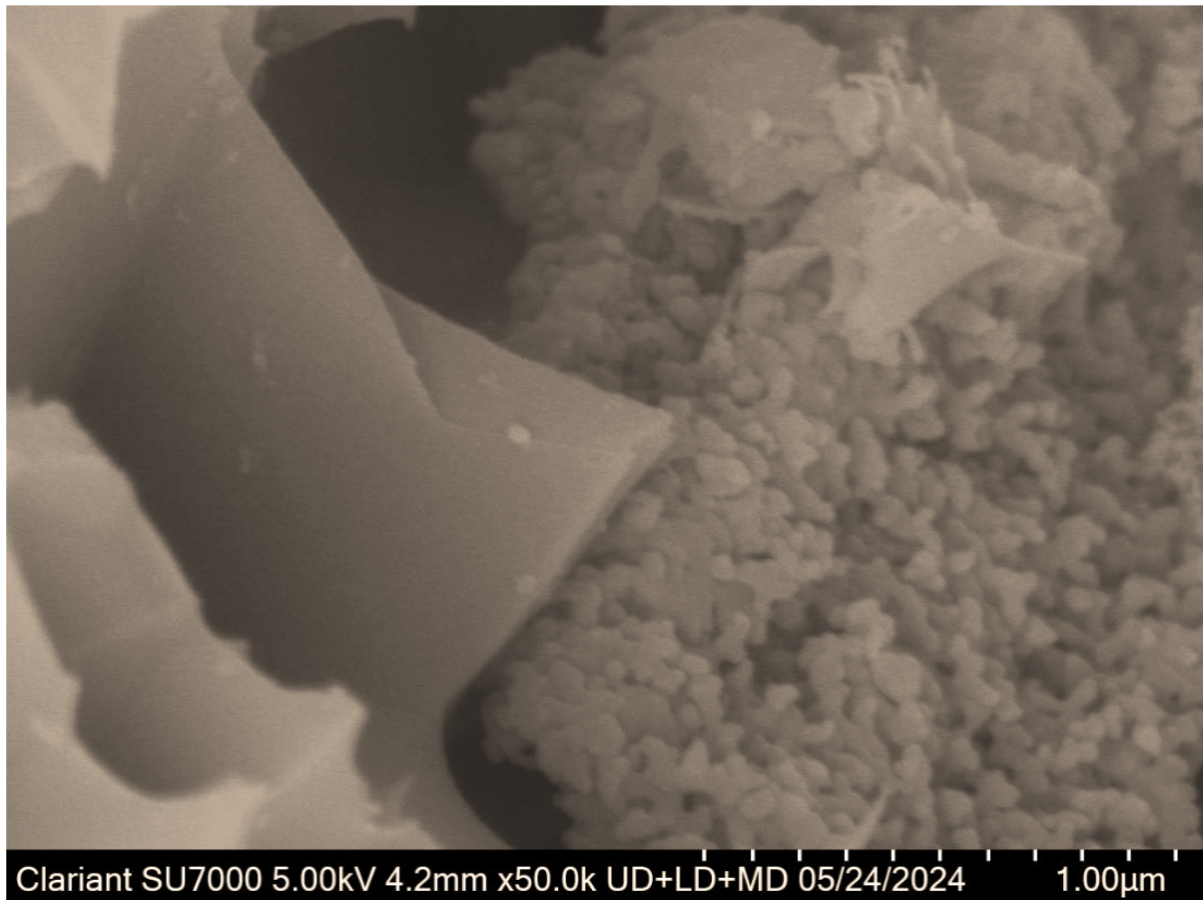






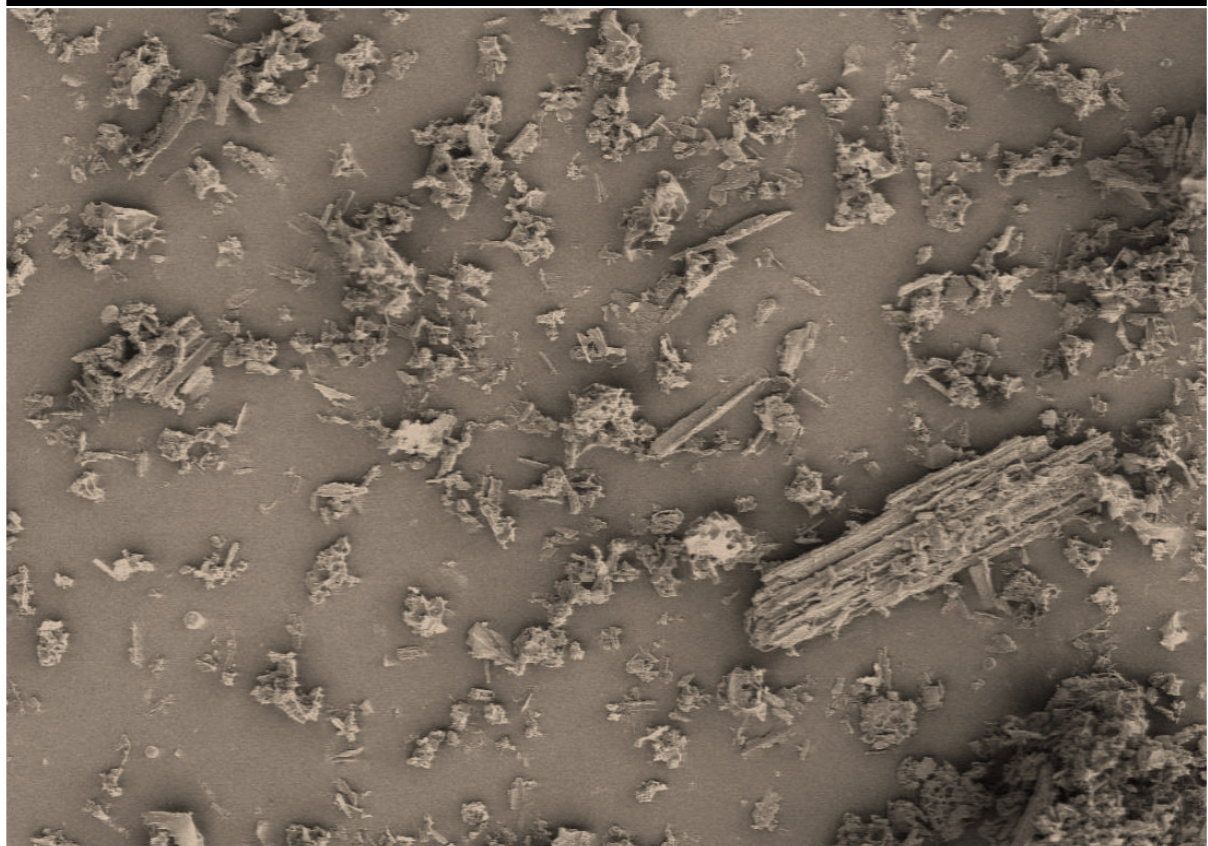






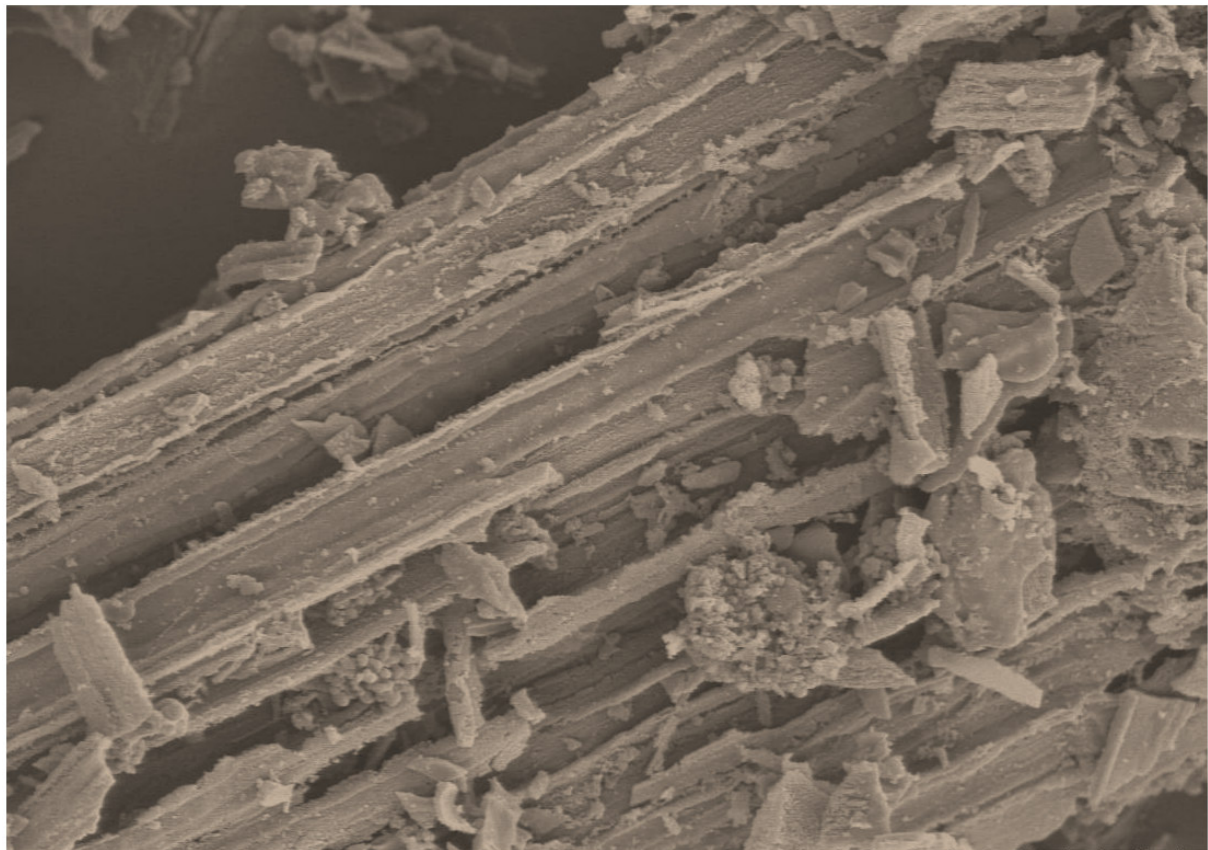


Clariant SU7000 5.00kV 4.2mm x100 UD+LD+MD 05/24/2024 500µm

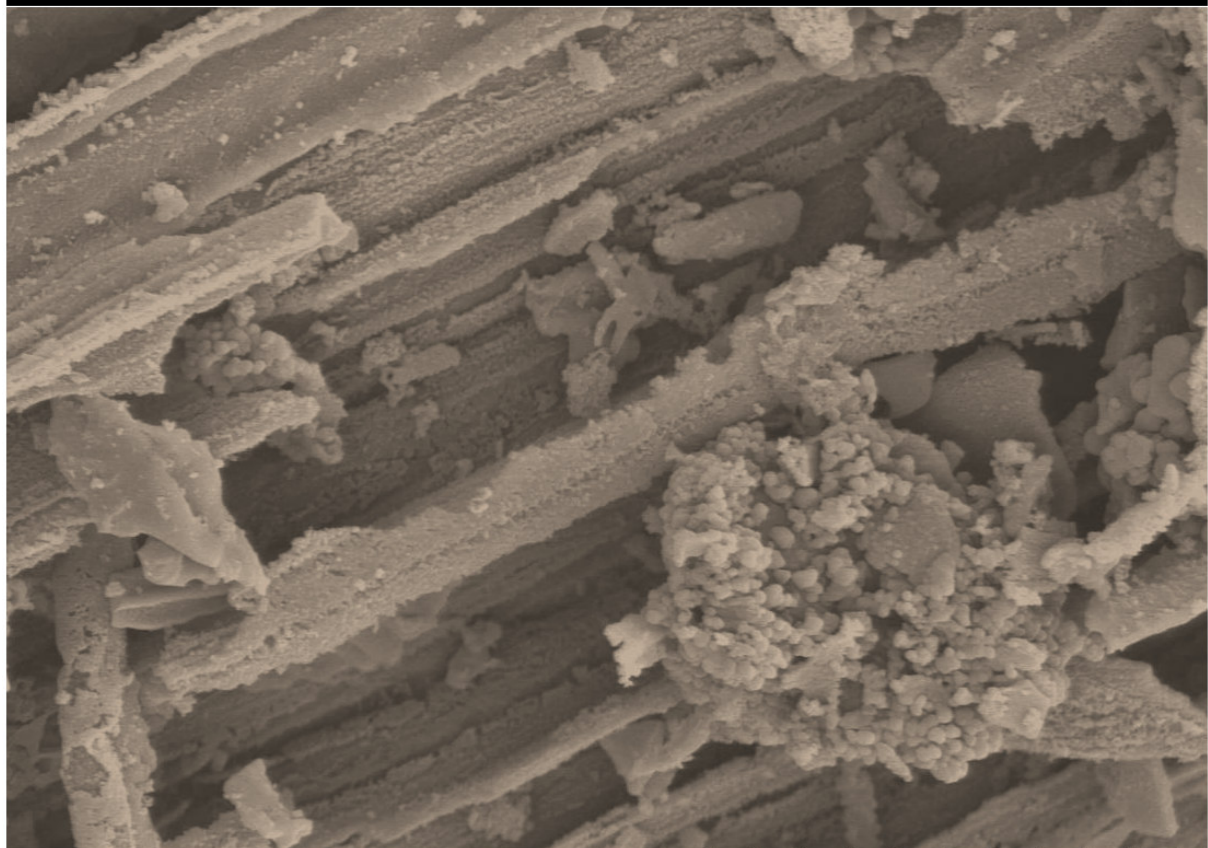


Clariant SU7000 5.00kV 4.3mm x200 UD+LD+MD 05/24/2024 200µm

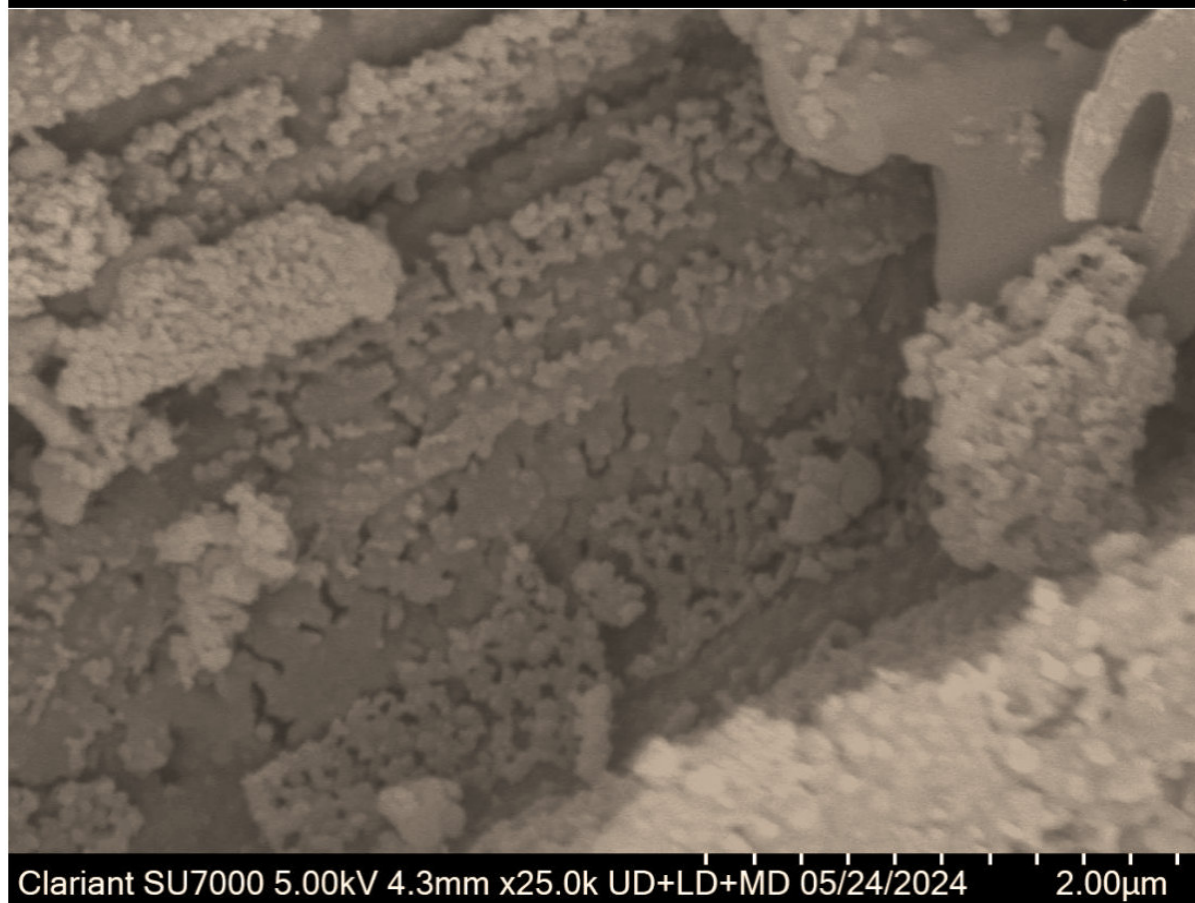
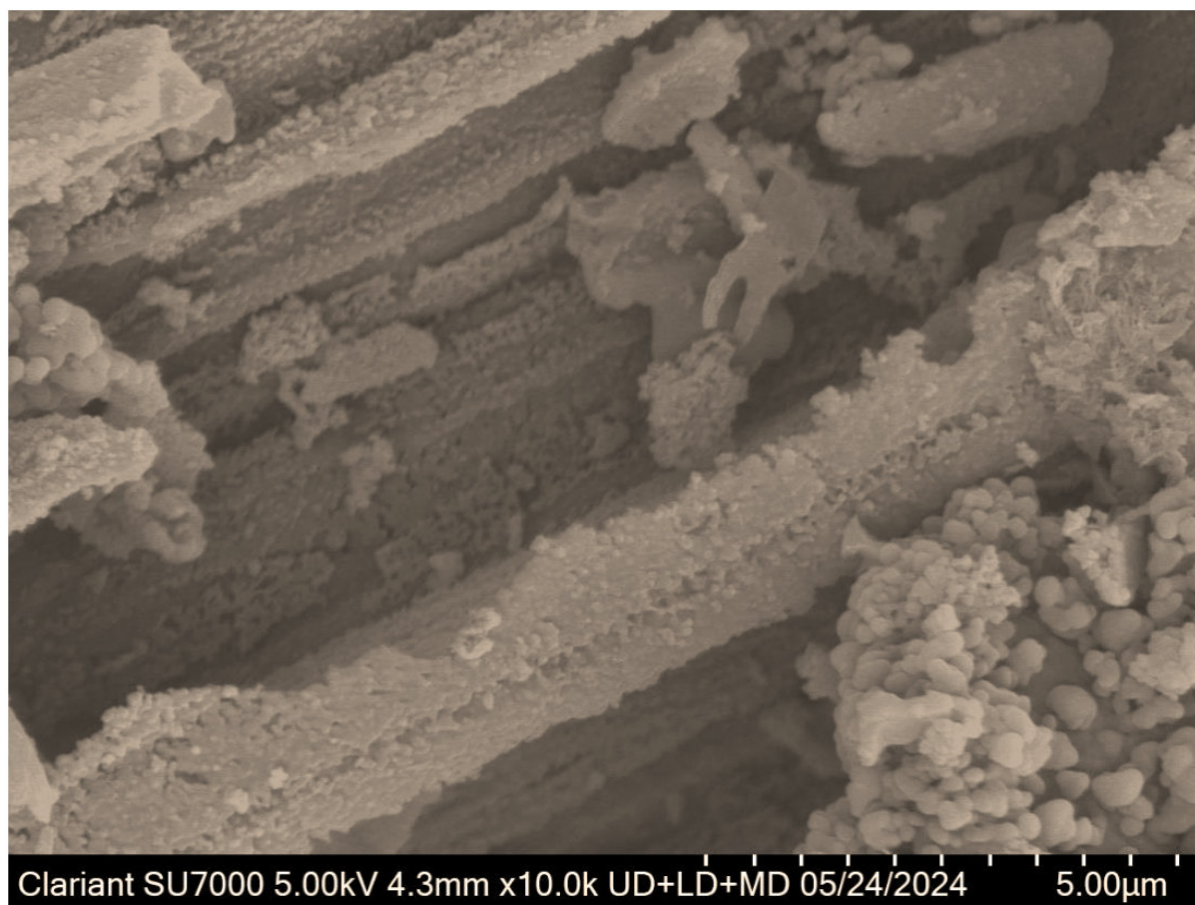




Clariant SU7000 5.00kV 4.3mm x2.00k UD+LD+MD 05/24/2024 20.0µm



Clariant SU7000 5.00kV 4.3mm x5.00k UD+LD+MD 05/24/2024 10.0µm



Experiment overview

Experiment	Reactor	Raw material	Acid	T	t	pressure	Wash	Output	Experiment	Qty	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output	
MB2928	AV1	6.0 g BAFA neu Hemp Fibre VF	100.0 g 0.19% 0.02 M Sulfuric acid	180 °C	22.5 h	8.7 bar -> 10.4 bar	742 g Water	2.6 g	->											
Comment	Fibres not completely covered by liquid																			
MB2933	AV1	6.1 g BAFA neu Hemp Fibre VF	174.0 g 0.19% 0.02 M Sulfuric acid	180 °C	24 h	8.7 bar -> 9.5 bar	647 g Water	1.2 g	->	TB1131	1.2 g	1.2 g KOH (90%)	manual	Ar	20-27 °C/min	790 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.16 g
Comment	dark crusts on glassware, metal and liquid surface removed before filtration									Comment	ground biochar: brown powder, strong electrostatic charge						(+ water)		(atm. Pressure)	
MB2935	AV1	6.0 g BAFA neu Hemp Fibre VF	172.9 g 0.19% 0.02 M Sulfuric acid	180 °C	23 h	8.8 bar -> 9.5 bar	500 g Water	1.3 g	->	TB1132	1.3 g	1.3 g KOH (90%)	mill (2 min)	Ar	20-29 °C/min	792 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.3 g
Comment	dark crusts on glassware, metal and liquid surface removed before filtration									Comment	ground biochar: brown powder						(+ water)		(atm. Pressure)	
MB2936	AV1	6.0 g BAFA neu Hemp Fibre VF	172.4 g 0.96% 0.10 M Sulfuric acid	180 °C	23 h	8.8 bar -> 10.6 bar	589 g Water	0.8 g		TB1133	1.0 g	1.0 g KOH (90%)	mill (1 min)	Ar	3 °C/min	785 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.3 g
Comment	dark crusts on glassware, metal and liquid surface removed before filtration									Comment	ground biochar: brown powder; see temperature trend						(+ water)		(atm. Pressure)	
MRa225	AV1	6.0 g BAFA neu Hemp Fibre VF	153.8 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	? -> 10.5 bar	500 g Water	0.9 g		TB1134	1.0 g	1.0 g KOH (90%)	mill (2 min)	Ar	3 °C/min	771 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.3 g
Comment	dark crusts on glassware, metal and liquid surface removed before filtration									Comment	ground biochar: brown powder; see temperature trend						(+ water)		(atm. Pressure)	
MRa228	AV1	6.3 g BAFA neu Hemp Fibre VF	154.0 g 0.96% 0.10 M Sulfuric acid	180 °C	22.5 h	8.7 bar -> 11.0 bar	500 g Water	0.9 g	->											
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																			
MRa231	AV1	6.1 g BAFA neu Hemp Fibre VF	168.8 g 0.96% 0.10 M Sulfuric acid	180 °C	23.5 h	8.0 bar -> 10.5 bar	500 g Water	0.8 g												
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																			
MB2937	AV1	9.4 g Canadian Rockies Hemp	179.3 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	8.6 bar -> 10.9 bar	569 g Water	1.6 g	->											
Comment	very low amount of foreign material observed and removed during filtration																			
MB2938	AV1	8.9 g Canadian Rockies Hemp	175.0 g 0.96% 0.10 M Sulfuric acid	180 °C	23.5 h	9.2 bar -> 13.2 bar	577 g Water	1.4 g	->											
Comment	very low amount of foreign material observed and removed during filtration																			
MB2940	AV1	9.2 g Canadian Rockies Hemp	177.1 g 0.96% 0.10 M Sulfuric acid	180 °C	22.5 h	8.7 bar -> 10.6 bar	555 g Water	1.5 g	->											
Comment	very low amount of foreign material observed and removed during filtration																			
MB2942	AV1	8.9 g Canadian Rockies Hemp	177.6 g 0.96% 0.10 M Sulfuric acid	180 °C	23 h	8.3 bar -> 13.5 bar	578 g Water	1.6 g	->											
Comment	low amount of foreign material observed and removed during filtration																			
MB2943	AV1	9.8 g Canadian Rockies Hemp	178.2 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	8.3 bar -> 12.7 bar	611 g Water	1.7 g	->											
Comment	low amount of foreign material observed and removed during filtration																			
MB2944	AV1	9.0 g Canadian Rockies Hemp	178.4 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	8.4 bar -> 9.7 bar	593 g Water	1.8 g	->											
Comment	low amount of foreign material observed and removed during filtration																			
MB2945	AV1	9.1 g Canadian Rockies Hemp	177.7 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	8.7 bar -> 9.8 bar	6 g Water	1.6 g	->											
Comment	low amount of foreign material observed and removed during filtration																			